

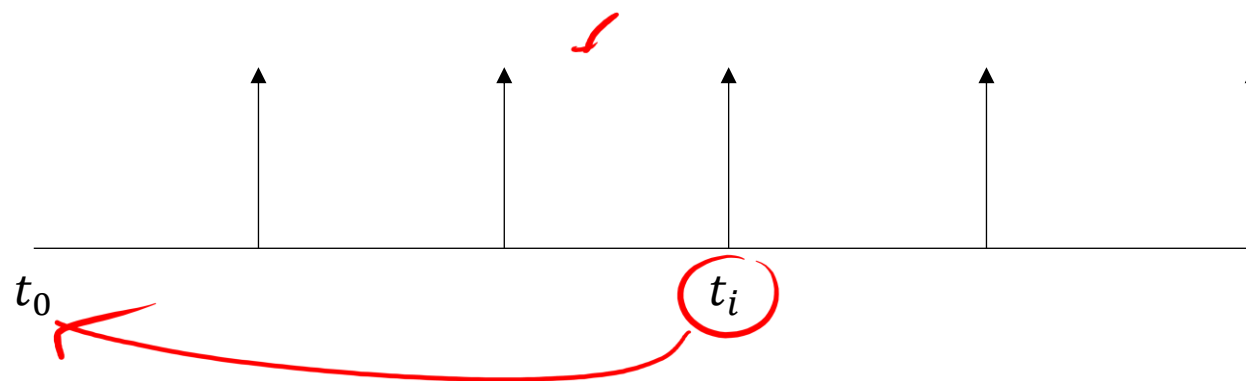
RFR Regime

(4. SOFR Swaps – Valuation & Sensitivities)

IRS Pricing - Annuity

- Given a stream of coupon payments, Annuity (A) is defined as the present value of unit coupons on unit Notional

$\sum (N \cdot r_i \cdot \tau_i \times P(t_0, t_i))$
 $A = \sum_{i=1}^n \tau_i P(t_0, t_i)$
 $A \times r_{\text{fixed}}$



- We will enhance this definition to include a notional amount N

$$A_N = N \sum_{i=1}^n \tau_i P(t_0, t_i)$$

(when Notional is constant)

$$A_{N_i} = \sum_{i=1}^n N_i \tau_i P(t_0, t_i)$$

(when Notional is variable)

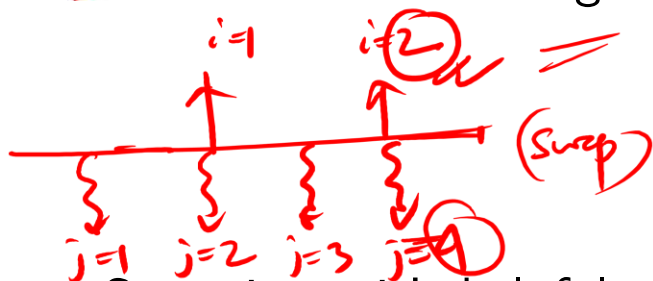
IRS Pricing – Value of Fixed/Floating Legs

► Value of the Fixed Leg

$$PV^{\text{Fixed Leg}} = r^{\text{Fixed}} \sum_{i=1}^n N_i \tau_i P(t_0, t_i) = r^{\text{Fixed}} \cdot A_{N_i}^{\text{Fixed}}$$

(Handwritten notes: i = 1 to n)

► Value of the Floating Leg



$$PV^{\text{Floating Leg}} = \sum_{j=1}^m N_j (r_j + s) \tau_j P(t_0, t_j)$$

(Handwritten notes: j = 1 to m, s=0)

Sometimes, it's helpful to bifurcate the floating and spread components

$$PV^{\text{Floating Leg}} = \underbrace{\sum_{j=1}^m N_j r_j \tau_j P(t_0, t_j)}_{\text{(floating)}} + s \underbrace{\sum_{j=1}^m N_j \tau_j P(t_0, t_j)}_{\text{(fixed)}} = \sum_{j=1}^m N_j r_j \tau_j P(t_0, t_j) + s A_{N_j}^{\text{Float}}$$

(Handwritten notes: s=0)

IRS Pricing – Par Rate

- ▶ The Fixed rate that makes the NPV of swap = 0 is called as the par rate. At inception typically swaps have 0 values and the fixed rate = par rate. Swaps that are executed with fixed rate = par rate are called par swaps.
- ▶ To find par rate, we make fixed leg = floating leg (without spread)

Handwritten notes and equations for IRS Pricing – Par Rate:

A_N (circled)

$N r^{\text{Fixed}} A^{\text{Fixed}} = N \sum_{j=1}^m r_j \tau_j P(t_0, t_j)$

$p = \frac{\sum_{j=1}^m r_j \tau_j P(t_0, t_j)}{A^{\text{Fixed}}}$

$p = \frac{PV_{s=0}^{\text{Floating Leg}}}{A_N^{\text{Fixed}}} = \frac{PV^{\text{Floating Leg}} - SA_N^{\text{Float}}}{A_N^{\text{Fixed}}}$

swap rate / par rate

$= \frac{1 - DF_N}{\sum_{j=1}^m \tau_j DF_j}$

$N \cdot r_{\text{fixed}} \cdot A^{\text{Fixed}} = N \sum_{j=1}^m \tau_j r_j P(t_0, t_j)$

$p = \frac{\sum_{j=1}^m \tau_j r_j P(t_0, t_j)}{N A^{\text{Fixed}}}$

$p = \frac{PV^{\text{Floating Leg}} - SA_N^{\text{Float}}}{A_N^{\text{Fixed}}}$

Receiver Swap and Payer Swap

- A receiver swap is the one that receives fixed and pays floating. Similarly, a payer swap is the one that pays fixed and receives floating. Let's define indicator ϕ to denote the position, so that

$$\phi = \begin{cases} 1, & \text{Receiver} \\ -1, & \text{Payer} \end{cases}$$

- Value of swap is given as

Handwritten notes:

- $V = \phi \cdot r^{\text{fixed}} \cdot A_N^{\text{fixed}}$ (with A_N^{fixed} circled)
- $\frac{dv}{dr} = \phi \cdot A_N^{\text{fixed}}$

Handwritten notes:

- $PV^{\text{fixed}} - PV^{\text{float}}$
- $PV^{\text{swap}} = \phi (PV^{\text{Fixed Leg}} - PV^{\text{Floating Leg}})$

$$PV^{\text{swap}} = \phi \left(r^{\text{Fixed}} A_N^{\text{Fixed}} - \sum_{j=1}^m N(r_j + s) \tau_j P(t_0, t_j) \right)$$

$$PV^{\text{swap}} = \phi \left[(r^{\text{Fixed}} - p) A_N^{\text{Fixed}} - s A_N^{\text{Float}} \right]$$

Handwritten notes:

- $P A_N^{\text{Fixed}}$
- $\left(\sum_{j=1}^m N \tau_j P(t_0, t_j) + s A_N^{\text{Floating}} \right)$

IRS Risk – Macaulay Duration

- **Macaulay Duration** : While Macaulay Duration is classically understood in the context of a bond, it's common market practice to calculate Duration for Fixed and Floating leg separately. There is no concept of a common yield for the swap.

For bonds we have

$$D_{Mac} = \sum_{i=1}^n t_i \left(\frac{\text{PV of } i\text{th Bond Coupon}}{\text{Bond Price}} \right)$$

$$\begin{aligned} CF_1 &\rightarrow \frac{1}{(1+y)^1} & \frac{1}{(1+r_1)} \\ CF_2 &\rightarrow \frac{1}{(1+y)^2} & \frac{1}{(1+r_2)} \end{aligned}$$

For fixed-leg of the swap, we have

$$D_{Mac}^{Fixed} = \sum_{i=1}^n t_i \left(\frac{\text{PV of } i\text{th Fixed Coupon}}{\text{PV of Fixed Leg}} \right) = \frac{r^{Fixed} \sum_{i=1}^n t_i N_i \tau_i P(t_0, t_i)}{r^{Fixed} \sum_{i=1}^n N_i \tau_i P(t_0, t_i)}$$

$$D_{Mac}^{Fixed} = \left(\frac{\sum_{i=1}^n t_i N_i \tau_i P(t_0, t_i)}{A_{N_t}^{Fixed}} \right)$$

IRS Risk – Macaulay Duration

- ▶ Similarly for floating-leg, we can calculate Macaulay Duration as

$$D_{Mac}^{Fixed} = \sum_{j=1}^m t_j \left(\frac{\text{PV of } j\text{th Floating Coupon}}{\text{PV of Floating Leg}} \right) = \frac{\sum_{j=1}^m t_j N_j r_j \tau_j P(t_0, t_j)}{\sum_{j=1}^m N_j r_j \tau_j P(t_0, t_j)}$$

- ▶ Spread duration can be calculated separately as it behaves like a fixed leg

$$D_{Mac}^{spread} = \left(\frac{\sum_{j=1}^m t_j N_j \tau_j P(t_0, t_i)}{A_{N_j}^{Spread}} \right)$$

IRS Risk – Modified Duration

- ▶ Modified duration in case of a bond refers to the % change in price because of change in yield. It's given as

$$\frac{D_{Mac}}{1 + \frac{y}{m}}$$

$$D_{mod} = \frac{D_{Mac}}{(1 + y)}$$

- ▶ But there is no such concept as a yield for a swap. However, when bond's coupon rate matches its yield, bond trades at par. Similarly, when a swap's fixed coupon rate = par rate the swap trades at par (value = 0). So, analogously we can treat the swap par rate as the yield.

$$D_{mod} = \frac{D_{Mac}}{(1 + p)}$$

Liquid instrument
 { Future Swaps } → 

IRS Risk – PV01

- In general, PV01 (present value of basis point) represents the sensitivity to 1 bp change in interest rates. In the context of swap, this refers to the sensitivity of the fixed leg to 1 bp increase in fixed rates. This metric ignores changes in discount factors

$$PV01^{swap} = (\phi A_{N_i}^{Fixed}) / 10000$$

While this is not accurate measure of sensitivity to yield curve, it's simple and fast to calculate analytically. A related concept is DV01 (next slide) which is more accurate. For par swaps PV01 is equal to DV01.



IRS Risk – DV01

- DV01 (dollar value of basis point) represents the sensitivity of the swap to 1 bp downward-shift in par rates. This includes changes in both forecast curve and discount curve making it a more comprehensive measure. Mathematically

$$DV01^{swap} = - \left(\frac{dPV^{swap}}{dp} \right) / 10000$$

Since $PV^{swap} = \phi \left[(r^{Fixed} - p) A_N^{Fixed} - s A_N^{Float} \right]$

$$\left(\frac{dPV^{swap}}{dp} \right) = -\phi \left(A_N^{Fixed} - (r^{Fixed} - p) \frac{dA_N^{Fixed}}{dp} - s \frac{dA_N^{Float}}{dp} \right)$$

After simplification, one obtains

$$DV01^{swap} = \phi \left(A_N^{Fixed} - (r^{Fixed} - p) A_N^{Fixed} D_{Mod}^{Fixed} - s A_N^{Float} D_{Mod}^{Float} \right) / 10000$$